1. Project Overview

The SSQ Attendance Management Module is designed to manage and analyze employee attendance across different teams, circles, projects, customers and work sites.

The project uses an attendance dataset containing **500 records** with employee information, team and project details, attendance dates, check-in/check-out timings, GPS information, attendance status, leave details, regularization status and approval information.

The main objective of the project is to provide a structured system for monitoring attendance, identifying attendance exceptions, analyzing employee and team attendance patterns, and presenting important attendance information through an interactive **Power BI dashboard**.

The project follows the workflow:

Attendance Data → Data Validation → PostgreSQL → SQL Analysis → Power BI → DAX Measures → Dashboard → Employee Drill-Through

2. Dataset Summary

The attendance dataset contains **500 attendance records** covering employee, team, project, site and attendance-related information.

Dataset Details

- **Number of Records:** 500
- **Number of Columns:** 23
- **Database:** PostgreSQL
- **Visualization Tool:** Microsoft Power BI
- **Analysis Language:** SQL and DAX

Key Features

The dataset contains the following major categories of information:

Employee Information

- Employee Code
- Employee Name
- Designation
- Department

- Reporting Manager

Team and Project Information

- Team Code
- Circle
- Project
- Customer
- Site ID

Attendance Information

- Attendance Date
- Check-In Time
- Check-Out Time
- Attendance Status
- Shift

Location Information

- Check-In GPS
- Check-Out GPS

Leave and Regularization

- Leave Type
- Regularization Status
- Approval Status
- Remarks

System Information

- Device/User ID
- Created/Modified Date & Time

Attendance Statuses

The dataset contains different attendance statuses including:

- Present

- Absent
- Leave
- Half Day
- Weekly Off
- Holiday

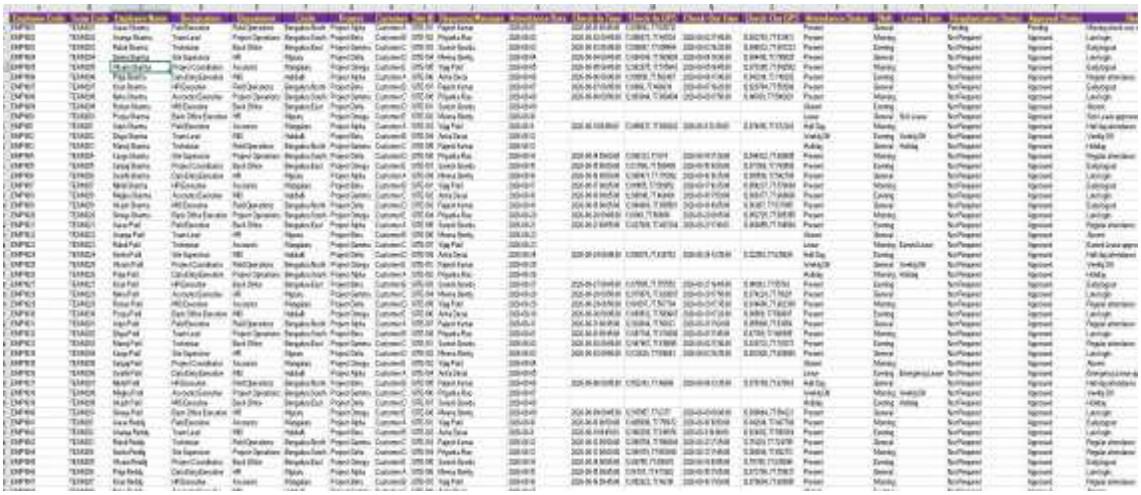


Figure 1: Attendance dataset structure

3. Data Preparation and Cleaning

Before performing analysis, the attendance data was validated and prepared for database and dashboard analysis.

The following data preparation activities were performed:

3.1 Missing Value Validation

Missing values were checked across important columns such as Check-In Time, Check-Out Time, GPS fields and Leave Type.

Not all blank values were treated as errors because some blanks are expected based on attendance status. For example, employees marked as **Absent, Leave, Weekly Off or Holiday** may not have check-in or check-out timings.

However, missing check-out values for employees marked as **Present or Half Day** were treated as attendance exceptions.

A total of **10 Present/Half Day records were identified with missing check-out values**.

3.2 Duplicate Validation

Attendance records were checked using **Employee Code and Attendance Date** to identify possible duplicate attendance entries.

3.3 Date and Time Validation

Attendance dates and check-in/check-out values were validated to ensure they were stored in appropriate date and timestamp formats.

3.4 Attendance Status Validation

Attendance status values were checked to ensure consistent categories such as Present, Absent, Leave, Half Day, Weekly Off and Holiday.

3.5 GPS Validation

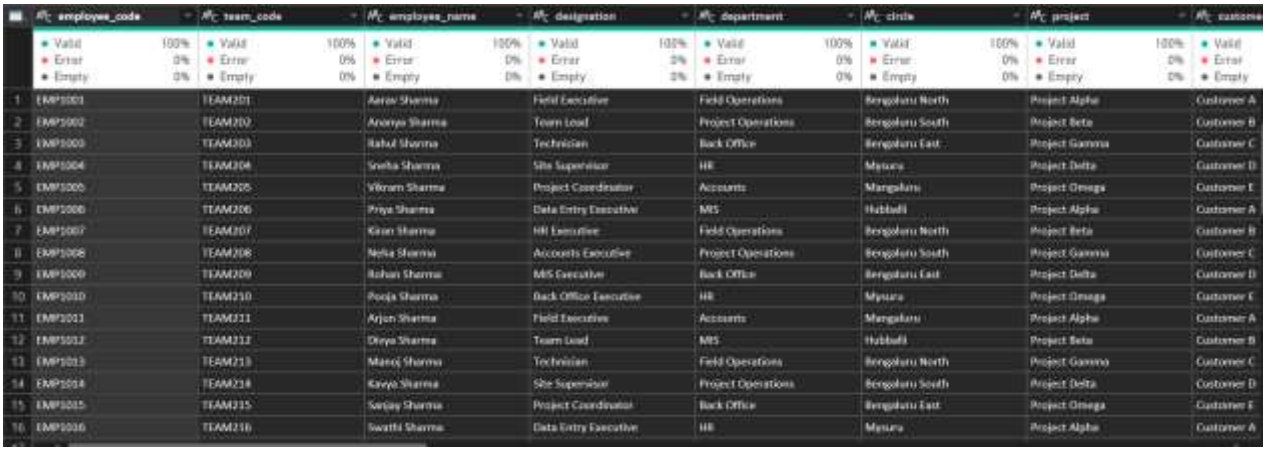
Check-in and check-out GPS fields were reviewed to identify missing or exceptional location information.

3.6 Exception Identification

Attendance exceptions such as:

- Late login
- Early logout
- Missing checkout
- Pending regularization
- Unapproved absence
- GPS-related exceptions

were identified for further analysis.



	employee_code	team_code	employee_name	designation	department	circle	project	customer
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	EMP1001	TEAM201	Aarav Sharma	Field Executive	Field Operations	Bengaluru North	Project Alpha	Customer A
2	EMP1002	TEAM202	Ananya Sharma	Team Lead	Project Operations	Bengaluru South	Project Beta	Customer B
3	EMP1003	TEAM203	Rahul Sharma	Technician	Back Office	Bengaluru East	Project Gamma	Customer C
4	EMP1004	TEAM204	Sneha Sharma	Site Supervisor	HR	Mysuru	Project Delta	Customer D
5	EMP1005	TEAM205	Vikram Sharma	Project Coordinator	Accounts	Mangaluru	Project Omega	Customer E
6	EMP1006	TEAM206	Priya Sharma	Data Entry Executive	MIS	Hubballi	Project Alpha	Customer A
7	EMP1007	TEAM207	Kiran Sharma	HR Executive	Field Operations	Bengaluru North	Project Beta	Customer B
8	EMP1008	TEAM208	Neha Sharma	Accounts Executive	Project Operations	Bengaluru South	Project Gamma	Customer C
9	EMP1009	TEAM209	Rohan Sharma	MIS Executive	Back Office	Bengaluru East	Project Delta	Customer D
10	EMP1010	TEAM210	Pooja Sharma	Back Office Executive	HR	Mysuru	Project Omega	Customer E
11	EMP1011	TEAM211	Arjun Sharma	Field Executive	Accounts	Mangaluru	Project Alpha	Customer A
12	EMP1012	TEAM212	Divya Sharma	Team Lead	MIS	Hubballi	Project Beta	Customer B
13	EMP1013	TEAM213	Manoj Sharma	Technician	Field Operations	Bengaluru North	Project Gamma	Customer C
14	EMP1014	TEAM214	Kavya Sharma	Site Supervisor	Project Operations	Bengaluru South	Project Delta	Customer D
15	EMP1015	TEAM215	Sanjay Sharma	Project Coordinator	Back Office	Bengaluru East	Project Omega	Customer E
16	EMP1016	TEAM216	Swathi Sharma	Data Entry Executive	HR	Mysuru	Project Alpha	Customer A

Figure 2: Data validation and missing-value analysis

4. Database Integration using PostgreSQL

After data preparation, the attendance dataset was imported into **PostgreSQL** for structured storage and SQL-based analysis.

An attendance table was created containing employee, team, project, attendance, GPS, leave and regularization information.

The database structure included fields such as:

- Employee Code
- Team Code
- Employee Name
- Designation
- Department
- Circle
- Project
- Customer
- Site ID
- Reporting Manager
- Attendance Date
- Check-In Time
- Check-In GPS
- Check-Out Time
- Check-Out GPS
- Attendance Status
- Shift
- Leave Type
- Regularization Status
- Approval Status

- Remarks
- Device/User ID
- Created/Modified Date & Time

The CSV dataset was then imported into PostgreSQL and validated by checking the total number of records.

The database contained **500 attendance records** after successful import.

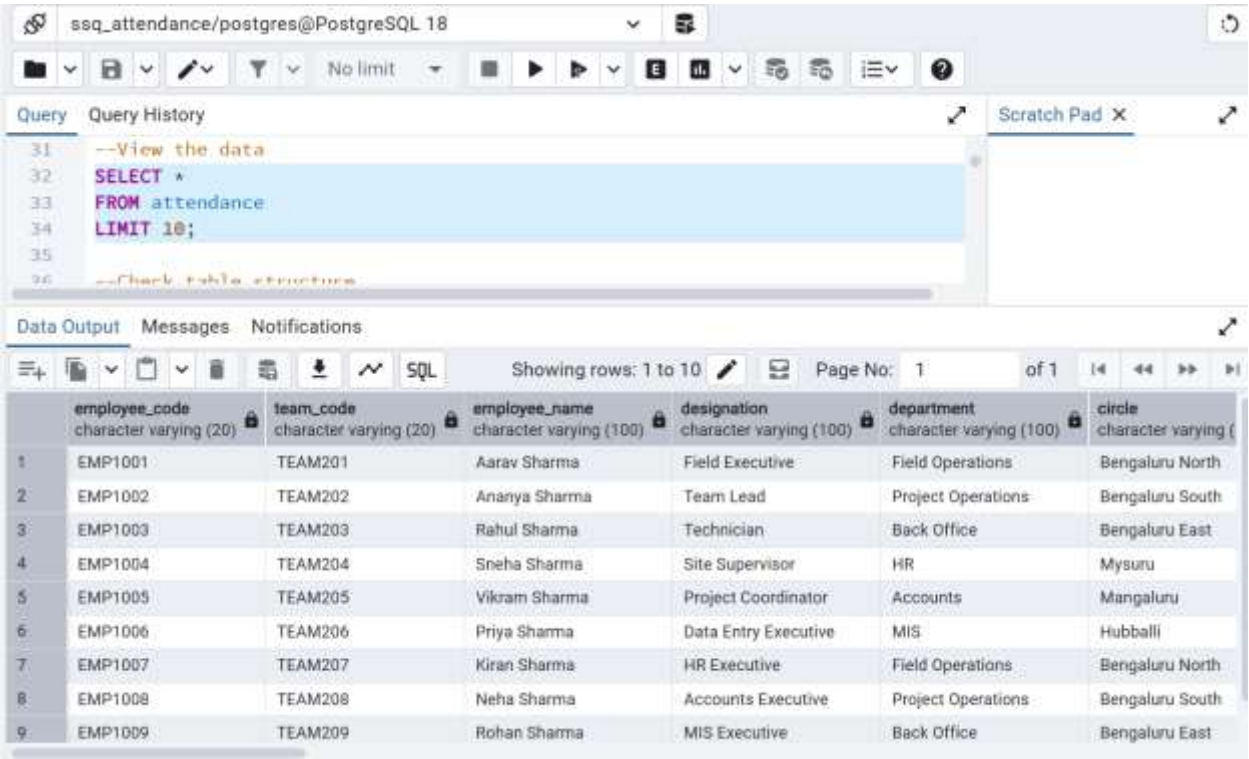


Figure 3: Attendance data imported into PostgreSQL

5. Data Analysis using SQL

SQL was used to analyze the attendance data and answer important operational questions.

The analysis was performed using PostgreSQL.

5.1 Total Attendance Records

The total number of attendance records was calculated.

Result: 500 records

This confirms that the complete dataset was successfully loaded and available for analysis.

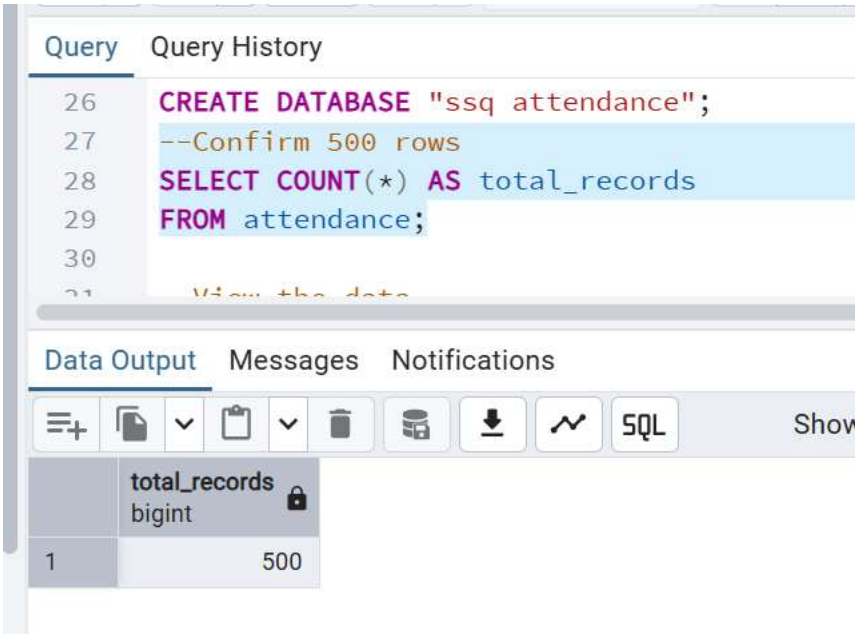


Figure 4: Total attendance records

5.2 Attendance Status Analysis

The number of records for each attendance status was calculated.

Attendance Status	Count
Present	310
Absent	38
Leave	38
Half Day	38
Weekly Off	38
Holiday	38

The analysis shows that **Present** is the largest attendance category in the dataset.

Query | Query History

```sql
--Check attendance status
SELECT
    attendance_status,
    COUNT(*) AS total_records
FROM attendance
GROUP BY attendance_status
ORDER BY total_records DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to

	attendance_status character varying (30)	total_records bigint
1	Present	310
2	Leave	38
3	Holiday	38
4	Half Day	38
5	Weekly Off	38
6	Absent	38

Figure 5: Attendance status distribution

5.3 Employee-wise Attendance Analysis

Employee-wise attendance was analyzed to understand individual attendance records and identify employees with attendance exceptions.

The analysis can be used to identify:

- Number of present days
- Absent days
- Leave days
- Half days
- Missing check-outs
- Attendance percentage

Query | Query History

```
81  --Employee-wise attendance
82  SELECT
83      employee_code,
84      employee_name,
85      COUNT(*) AS total_records,
86      COUNT(*) FILTER (
87          WHERE attendance_status = 'Present'
88      ) AS present_days,
89      COUNT(*) FILTER (
90          WHERE attendance_status = 'Absent'
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1

	employee_code character varying (20)	employee_name character varying (100)	total_records bigint	present_days bigint	absent_days bigint	leave_days bigint
1	EMP1001	Arjun Reddy	1	1	0	0
2	EMP1001	Swathi Joshi	1	0	0	0
3	EMP1001	Priya Gowda	1	0	1	0
4	EMP1001	Aarav Sharma	1	1	0	0
5	EMP1002	Divya Reddy	1	1	0	0
6	EMP1002	Kiran Gowda	1	0	0	1
7	EMP1002	Nikhil Joshi	1	0	0	0

Figure 6: Employee-wise attendance analysis

5.4 Team-wise Attendance Analysis

Attendance records were grouped by Team Code to compare attendance patterns across different teams.

This helps management identify teams with:

- Higher absence
- More leave records
- More attendance exceptions
- Missing punch issues

Query | Query History

```
119  --Team-wise attendance
120  SELECT
121      team_code,
122      COUNT(*) AS total_records,
123      COUNT(*) FILTER (
124          WHERE attendance_status = 'Present'
125      ) AS present_count,
126      COUNT(*) FILTER (
127          WHERE attendance_status = 'Absent'
128      ) AS absent_count,
129      COUNT(*) FILTER (
```

Data Output | Messages | Notifications

Showing rows: 1 to 30 | Page No: 1

	team_code character varying (20)	total_records bigint	present_count bigint	absent_count bigint	leave_count bigint	half_day_count bigint
1	TEAM201	17	10	2	1	1
2	TEAM202	17	11	1	2	1
3	TEAM203	17	11	1	1	2
4	TEAM204	17	11	1	1	1
5	TEAM205	17	10	2	1	1
6	TEAM206	17	11	1	2	1

Figure 7: Team-wise attendance analysis

5.5 Circle-wise Attendance Analysis

Attendance was analyzed based on Circle to understand attendance distribution across different operational areas.

This provides management with a higher-level view of attendance performance across circles.

Query Query History

```
140  --Circle-wise attendance
141  SELECT
142      circle,
143      attendance_status,
144      COUNT(*) AS total_records
145  FROM attendance
146  GROUP BY circle, attendance_status
147  ORDER BY circle, attendance_status;
```

Data Output Messages Notifications

Showing rows: 1 to 36

	circle character varying (100)	attendance_status character varying (30)	total_records bigint
1	Bengaluru East	Absent	7
2	Bengaluru East	Half Day	6
3	Bengaluru East	Holiday	6
4	Bengaluru East	Leave	6
5	Bengaluru East	Present	52
6	Bengaluru East	Weekly Off	6
7	Bengaluru North	Absent	6

Figure 8: Circle-wise attendance analysis

5.6 Project-wise Attendance Analysis

Attendance records were grouped by Project to understand attendance patterns across different projects.

This analysis can help project managers monitor workforce availability and identify projects with attendance exceptions.

```sql
--Project-wise attendance
SELECT
    project,
    attendance_status,
    COUNT(*) AS total_records
FROM attendance
GROUP BY project, attendance_status
ORDER BY project, attendance_status;
```

Data Output | Messages | Notifications

Showing rows: 1 to 30

	project character varying (100)	attendance_status character varying (30)	total_records bigint
1	Project Alpha	Absent	7
2	Project Alpha	Half Day	8
3	Project Alpha	Holiday	8
4	Project Alpha	Leave	8
5	Project Alpha	Present	62
6	Project Alpha	Weekly Off	7
7	Project Beta	Absent	8

Figure 9: Project-wise attendance analysis

5.7 Missing Check-Out Analysis

Records with missing check-out times were identified.

A total of **10 Present/Half Day records** were identified with missing check-out values.

These records represent potential attendance exceptions that may require employee regularization or manager verification.

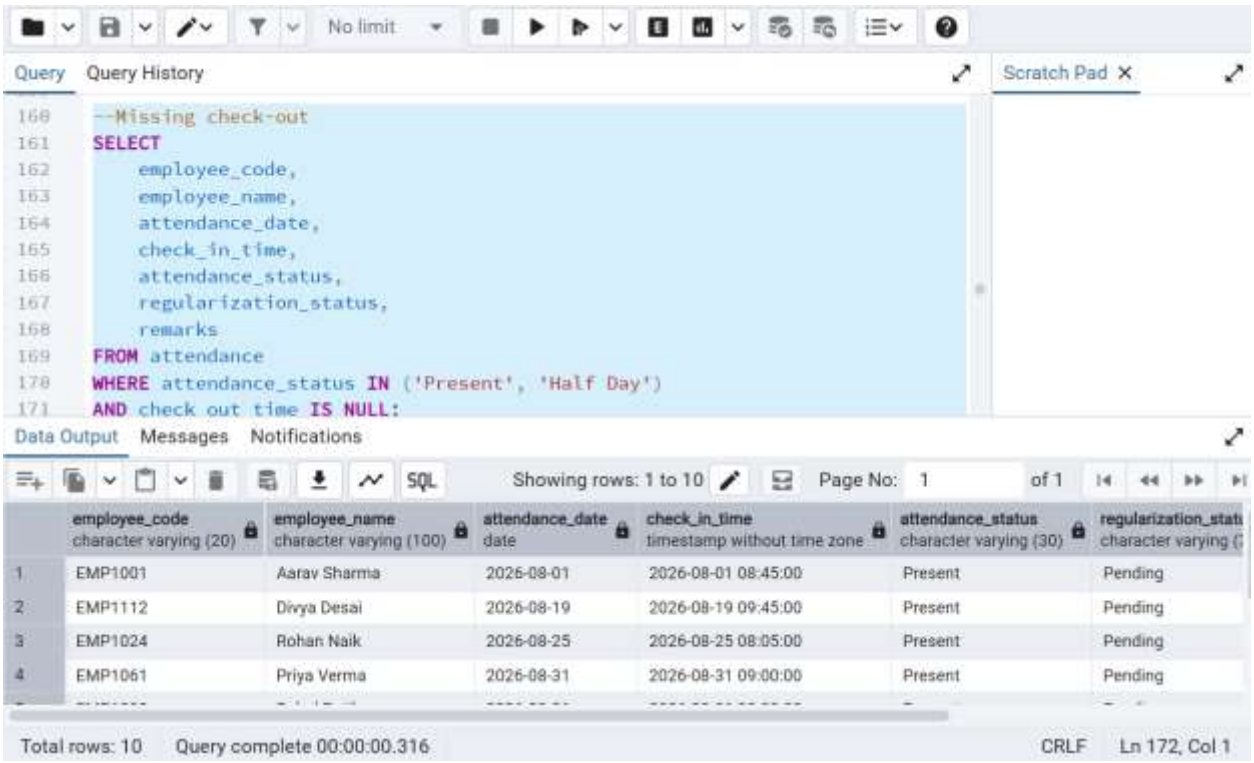
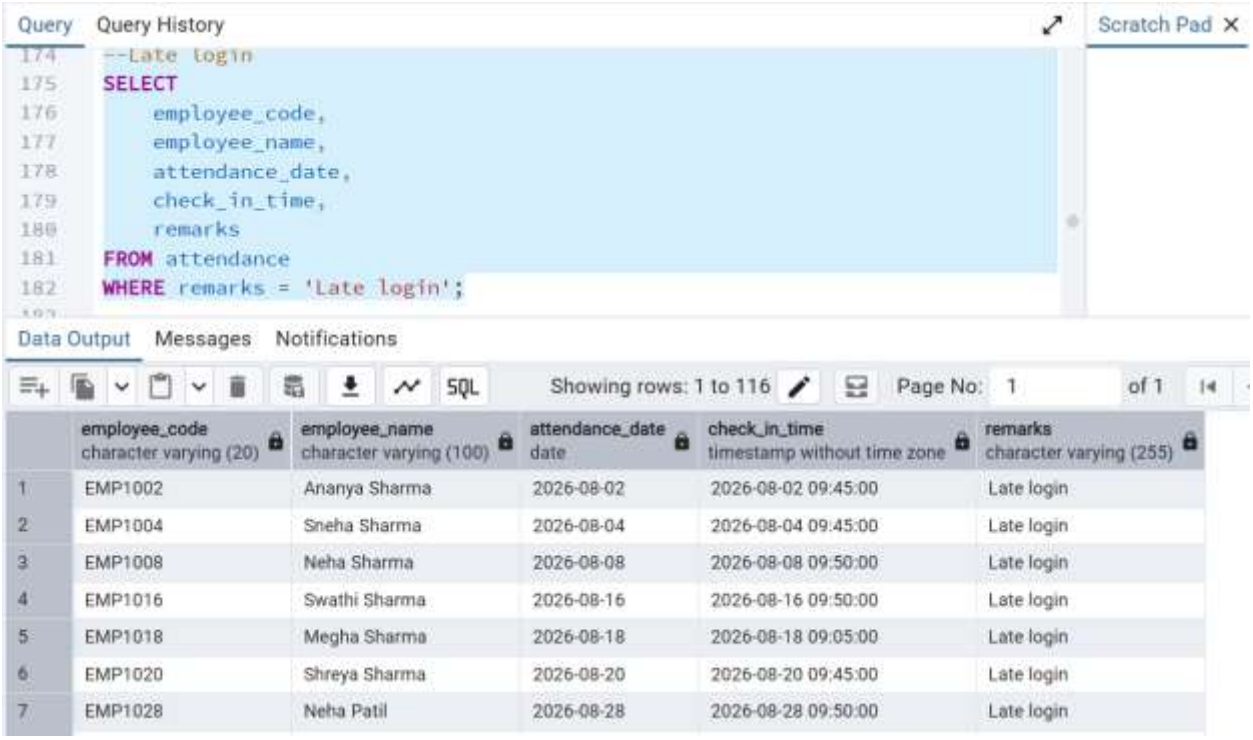


Figure 10: Employees with missing check-out

5.8 Late Login Analysis

Employees with check-in times later than the defined expected login time were identified.

This analysis helps management monitor late attendance and identify repeated late-login patterns.



```
Query  Query History                                                    Scratch Pad ×
174  --Late login
175  SELECT
176      employee_code,
177      employee_name,
178      attendance_date,
179      check_in_time,
180      remarks
181  FROM attendance
182  WHERE remarks = 'Late login';
```

Data Output | Messages | Notifications

Showing rows: 1 to 116 | Page No: 1 of 1

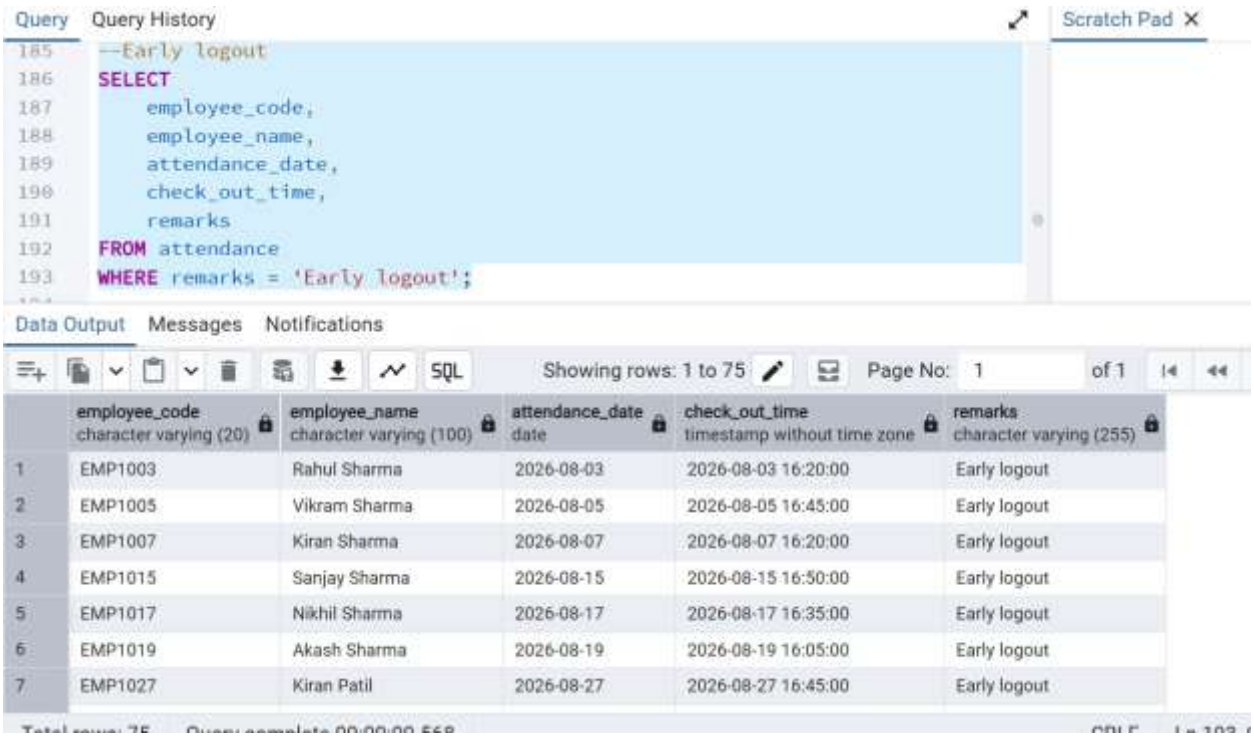
	employee_code character varying (20)	employee_name character varying (100)	attendance_date date	check_in_time timestamp without time zone	remarks character varying (255)
1	EMP1002	Ananya Sharma	2026-08-02	2026-08-02 09:45:00	Late login
2	EMP1004	Sneha Sharma	2026-08-04	2026-08-04 09:45:00	Late login
3	EMP1008	Neha Sharma	2026-08-08	2026-08-08 09:50:00	Late login
4	EMP1016	Swathi Sharma	2026-08-16	2026-08-16 09:50:00	Late login
5	EMP1018	Megha Sharma	2026-08-18	2026-08-18 09:05:00	Late login
6	EMP1020	Shreya Sharma	2026-08-20	2026-08-20 09:45:00	Late login
7	EMP1028	Neha Patil	2026-08-28	2026-08-28 09:50:00	Late login

Figure 11: Late login analysis

5.9 Early Logout Analysis

Attendance records were analyzed to identify employees who checked out earlier than the expected working time.

This provides an additional exception category for attendance verification.



```
--Early logout
SELECT
    employee_code,
    employee_name,
    attendance_date,
    check_out_time,
    remarks
FROM attendance
WHERE remarks = 'Early logout';
```

Showing rows: 1 to 75 | Page No: 1 of 1

	employee_code character varying (20)	employee_name character varying (100)	attendance_date date	check_out_time timestamp without time zone	remarks character varying (255)
1	EMP1003	Rahul Sharma	2026-08-03	2026-08-03 16:20:00	Early logout
2	EMP1005	Vikram Sharma	2026-08-05	2026-08-05 16:45:00	Early logout
3	EMP1007	Kiran Sharma	2026-08-07	2026-08-07 16:20:00	Early logout
4	EMP1015	Sanjay Sharma	2026-08-15	2026-08-15 16:50:00	Early logout
5	EMP1017	Nikhil Sharma	2026-08-17	2026-08-17 16:35:00	Early logout
6	EMP1019	Akash Sharma	2026-08-19	2026-08-19 16:05:00	Early logout
7	EMP1027	Kiran Patil	2026-08-27	2026-08-27 16:45:00	Early logout

Figure 12: Early logout analysis

5.10 Regularization Analysis

Regularization records were analyzed to identify attendance corrections that were pending or approved.

This helps the organization monitor attendance correction requests before attendance closure and payroll processing.

6. Power BI Data Preparation

The PostgreSQL attendance table was connected to **Microsoft Power BI** for interactive visualization.

The public attendance table was loaded into Power BI.

Power Query was used to validate:

- Employee Code
- Team Code
- Employee Name

- Attendance Date
- Attendance Status
- Check-In Time
- Check-Out Time
- GPS information
- Leave Type
- Regularization information

The data was then applied to the Power BI model for dashboard development.

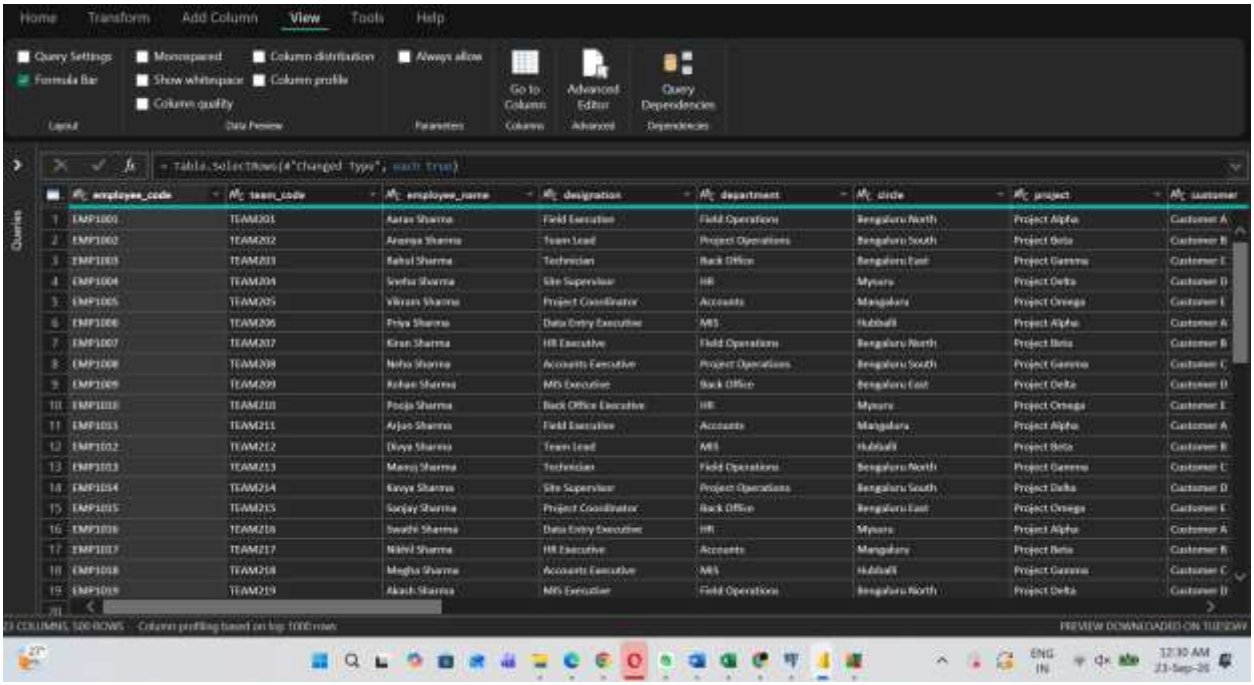


Figure 13: Attendance data preparation in Power BI

7. DAX Measures

DAX measures were created in Power BI to calculate important attendance KPIs.

Total Records:Result: 500

```
1 Total Records = COUNTROWS('public attendance')
```

Present Count : Result: 310

```
Present Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Present"
)
```

Absent Count : Result: 38

```
Absent Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Absent"
)
```

Leave Count : Result: 38

```
Leave Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Leave"
)
```

Half Day Count :Result: 38

```
Half Day Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Half Day"
)
```

Weekly Off Count :Result: 38

```
Weekly Off Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Weekly Off"
)
```

Holiday Count :Result: 38

```
Holiday Count =
CALCULATE(
    COUNTROWS('public attendance'),
    'public attendance'[attendance_status] = "Holiday"
)
```

Attendance Percentage : Result: 62.00%

```
Attendance % =
DIVIDE(
    [Present Count],
    [Total Records],
    0
) * 100
```

8. Power BI Dashboard

An interactive attendance dashboard was developed using Microsoft Power BI.

The dashboard provides a consolidated view of employee attendance and operational exceptions.

Key Performance Indicators

The dashboard displays:

- Total Records
- Present Count
- Absent Count
- Leave Count
- Half Day Count
- Weekly Off Count
- Holiday Count
- Attendance Percentage
- Late Login Count
- Missing Check-Out

Visual Analysis

The dashboard contains visualizations for:

- Attendance status distribution
- Team-wise attendance
- Project-wise attendance
- Circle-wise attendance
- Daily attendance trends

- Attendance exceptions

Interactive filters/slicers allow users to analyze the data based on relevant dimensions such as employee, team, project, status and date/month.



Figure 14: SSQ Attendance Management Power BI Dashboard

9. Employee Drill-Through Analysis

An employee-level drill-through page was created to provide detailed attendance information for an individual employee.

From the main dashboard, an employee can be selected and the drill-through option can be used to navigate to the employee detail page.

The detailed page provides the employee's attendance records, including:

- Attendance Date
- Check-In Time
- Check-Out Time
- Attendance Status
- Leave Information

- Regularization Information
- Project/Team details

This allows managers or HR personnel to move from a high-level dashboard view to individual attendance records.

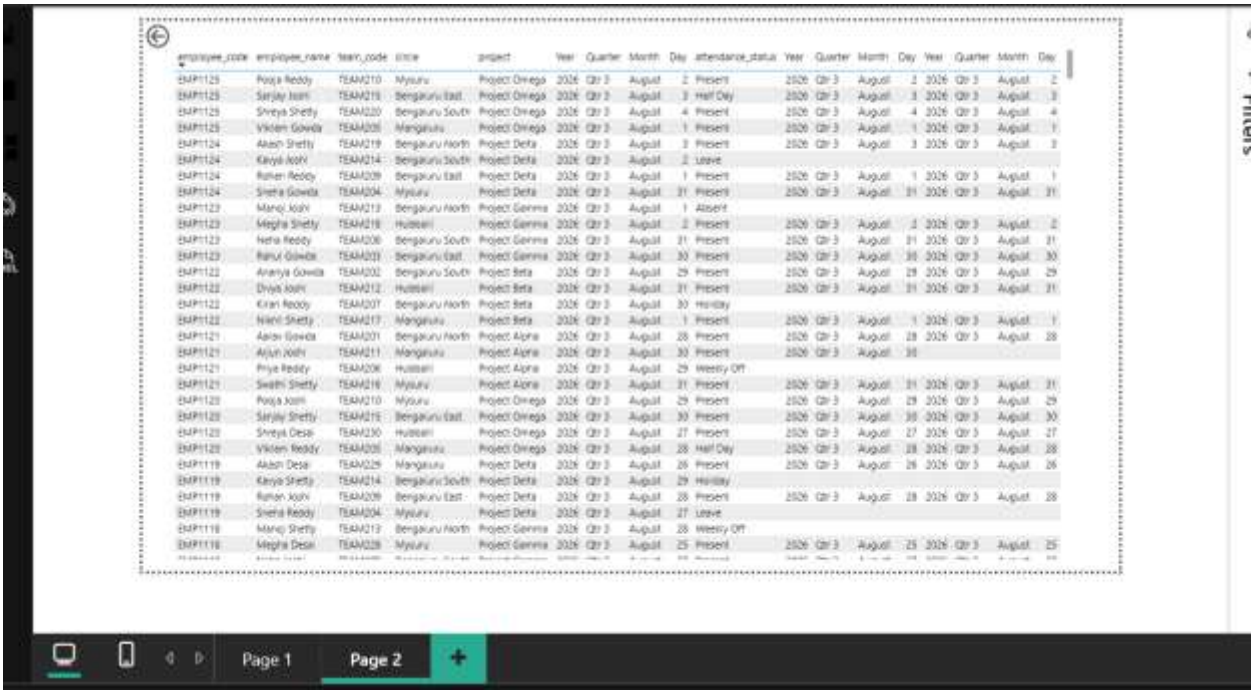


Figure 15: Employee attendance drill-through page

10. Key Findings

The major findings from the attendance analysis are:

1. The dataset contains **500 attendance records**.
2. **310 records are marked as Present**, making Present the largest attendance category.
3. Absent, Leave, Half Day, Weekly Off and Holiday each contain **38 records**.
4. The calculated attendance percentage is 62.00%, based on Present records divided by total attendance records.
5. **10 Present/Half Day records have missing check-out times**, which can be treated as attendance exceptions requiring verification.

6. Late login records were identified through SQL analysis and included in the Power BI dashboard.
7. Employee-wise, team-wise, circle-wise and project-wise attendance analysis provides multiple levels of attendance monitoring.
8. Regularization information can be used to track attendance corrections and pending approvals.

11. Business Benefits

The SSQ Attendance Management Module provides the following benefits:

Centralized attendance data management

- Faster attendance monitoring
- Employee-level attendance visibility
- Team and project-level attendance analysis
- Identification of missing punches
- Monitoring of late login and early logout
- Better visibility of leave and regularization
- GPS-related attendance validation
- Reduced manual attendance tracking
- Improved reporting for HR and management
- Better preparation for attendance closure and payroll-related review

12. Business Recommendations

Based on the attendance analysis, the following actions can be considered:

- **Monitor Missing Check-Outs:** Employees with missing check-outs should be verified before attendance closure.
- **Track Late Login:** Repeated late-login records can be reviewed by reporting managers.
- **Regularization Follow-up:** Pending regularization requests should be reviewed and approved within the required timeline.

- **Monitor Absence:** Unapproved absence records can be escalated to the appropriate manager.
- **Use Team and Project Dashboards:** Project managers can monitor workforce availability at team and project levels.
- **Strengthen GPS Validation:** GPS information can be used to validate field attendance and identify location-related exceptions.
- **Use Drill-Through for Employee Review:** HR and managers can investigate individual attendance records directly from the dashboard.

13. Conclusion

The SSQ Attendance Management Module provides a structured approach to managing, analyzing and monitoring employee attendance.

The project combines **PostgreSQL for data storage and SQL analysis**, **Power BI for visualization**, and **DAX for attendance KPI calculations**.

The module provides visibility into attendance status, employee and team attendance, project-wise attendance, late login, missing checkout, leave and regularization information.

The Power BI dashboard converts the attendance data into an interactive reporting solution that can support HR, project managers and management in monitoring attendance and identifying exceptions.

Overall, the project demonstrates how attendance data can be structured, analyzed and visualized using PostgreSQL, SQL and Power BI to support attendance monitoring and operational decision-making.

14. Appendix – Technologies Used

Technology	Purpose
CSV / Excel	Source attendance dataset
PostgreSQL	Database storage
SQL	Data analysis
Power Query	Data preparation and validation
Power BI	Dashboard and visualization

Technology	Purpose
DAX	KPI and attendance calculations



Figure 16: Final SSQ Attendance Management reporting solution